Initial Project Planning Template

Date	17 June 2025
Team ID	SWTXT_20251747173488
Project Name	Productivity Prediction Web App
Maximum Marks	4 Marks

Product Backlog, Sprint Schedule, and Estimation

Project Planning: Productivity Prediction Web App

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Introduction

The Productivity Prediction Web App leverages machine learning to forecast garment worker productivity using the "Productivity Prediction of Garment Employees" dataset. This planning document outlines the product backlog and sprint schedule for developing the app, which employs Linear Regression ($R^2 = 0.197$), Random Forest ($R^2 = 0.547$), and XGBoost ($R^2 = 0.482$) models. The plan assumes a 6-week Agile project with three 2-week sprints, prioritizing data preprocessing, model training, and evaluation, followed by a web interface and deployment.

Product Backlog

The product backlog lists prioritized tasks required to complete the app, with effort estimated in story points (Fibonacci scale: 1, 2, 3, 5, 8).

Problem Backlog ID | Description | Priority | Estimation (Story Points)

PB-1 | Load and explore "garments_worker_productivity.csv" dataset | High | 3

PB-2 | Preprocess data (handle missing "wip", encode categorical features, derive "month") | High | 5

PB-3 | Train and evaluate Linear Regression model | High | 3

PB-4 | Train and evaluate Random Forest model | High | 5

PB-5 | Train and evaluate XGBoost model | High | 5

PB-6 | Compare model performance (R^2, MAE, MSE) and save models | High | 3

PB-7 | Develop web interface for user inputs (Flask templates: predict, submit) | Medium | 5

PB-8 | Integrate ML models into Flask app for real-time predictions | Medium | 5

PB-9 | Create home and about pages for user navigation | Low | 2

PB-10 | Deploy app on Render with GitHub integration | Medium | 3

PB-11 | Write README and documentation (dataset, usage, results) | Medium | 3

Sprint Schedule

The project is divided into three 2-week sprints, balancing ML development with web interface and deployment tasks.

Sprint 1: Data and Model Development (June 16–19, 2025)

- PB-1: Load and explore dataset (3 points)

- PB-2: Preprocess data (5 points)

- PB-3: Train Linear Regression model (3 points)

- Total: 11 story points

Sprint 2: Advanced Models and Evaluation (June 20–June 22, 2025)

- PB-4: Train Random Forest model (5 points)

- PB-5: Train XGBoost model (5 points)

- PB-6: Compare model performance and save models (3 points)

- Total: 13 story points

Sprint 3: Web Interface and Deployment (June 23–26, 2025)

- PB-7: Develop web interface (5 points)

- PB-8: Integrate ML models into Flask app (5 points)

- PB-9: Create home and about pages (2 points)

- PB-10: Deploy on Render (3 points)

- PB-11: Write README and documentation (3 points)

- Total: 18 story points

Estimation Notes

- Story points reflect task complexity and effort, with ML tasks (e.g., preprocessing, model training) assigned higher points due to data handling and evaluation needs.

- Flask and Render tasks are estimated conservatively, assuming basic setup and integration.

- Total effort: 42 story points across 6 weeks, averaging 14 points per sprint, suitable for a solo developer or small team.